

Module 04 Lab - Exploratory Data Analysis **Objective:**

- To learn how to explore a dataset to find patterns, anomalies, and insights before modeling. EDA is like being a detective; you are looking for clues in the data that will help you build a better model. **This lab is fully coded.** Your task is to run each cell, read the detailed explanations, understand the purpose of each visualization, and then complete the experimentation section.

Part 1: Setup and Data Loading**What is Exploratory Data

Analysis (EDA)?**EDA is the process of using summary

statistics and visualizations to understand a dataset's main

characteristics. Before you can build a model, you need to

understand your data. What stories does it tell? Are there

errors or missing values? Are there strong relationships

between variables? EDA helps answer these questions. We will

use the famous Titanic dataset for this lab. It contains

information about passengers and, crucially, whether they

survived the disaster.

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns # Seaborn is a library built on top of Matplotlib that ma

# Load the dataset directly from a URL
df = pd.read_csv('https://raw.githubusercontent.com/datasciencedojo/datasets/m
```

```

print("--- First 5 Rows ---")
print(df.head())

print("--- Basic Info ---")
# .info() is a great first command. It tells us the column names, how many non-
# Notice that 'Age' and 'Cabin' have missing values!
df.info()

```

```

--- First 5 Rows ---

```

	PassengerId	Survived	Pclass	\
0	1	0	3	
1	2	1	1	
2	3	1	3	
3	4	1	1	
4	5	0	3	

	Name	Sex	Age	SibSp	\
0	Braund, Mr. Owen Harris	male	22.0	1	
1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	
2	Heikkinen, Miss. Laina	female	26.0	0	
3	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	
4	Allen, Mr. William Henry	male	35.0	0	

	Parch	Ticket	Fare	Cabin	Embarked
0	0	A/5 21171	7.2500	NaN	S
1	0	PC 17599	71.2833	C85	C
2	0	STON/O2. 3101282	7.9250	NaN	S
3	0	113803	53.1000	C123	S
4	0	373450	8.0500	NaN	S

```

--- Basic Info ---

```

```

<class 'pandas.core.frame.DataFrame'>

```

```

RangeIndex: 891 entries, 0 to 890

```

```

Data columns (total 12 columns):

```

#	Column	Non-Null Count	Dtype
0	PassengerId	891 non-null	int64
1	Survived	891 non-null	int64
2	Pclass	891 non-null	int64
3	Name	891 non-null	object
4	Sex	891 non-null	object
5	Age	714 non-null	float64
6	SibSp	891 non-null	int64
7	Parch	891 non-null	int64
8	Ticket	891 non-null	object
9	Fare	891 non-null	float64
10	Cabin	204 non-null	object
11	Embarked	889 non-null	object

```

dtypes: float64(2), int64(5), object(5)

```

```

memory usage: 83.7+ KB

```

- ✓ Part 2: Descriptive Statistics Let's start by getting a high-level numerical summary of the data. The `.describe()` method is perfect for this. It calculates statistics like mean, standard deviation, min, and max for the numerical columns.

```
# Get summary statistics for numerical columnsprint("--- Descriptive Statistics
```

Part 3: Visual EDA - Telling Stories with Plots Numbers are great, but plots make patterns and relationships immediately obvious. The goal of visual EDA is to turn data into insights.

- ✓ **Note on Plotting Libraries:**
 - * **Matplotlib:** The foundational library, gives you full control over everything.
 - * **Seaborn:** Built on Matplotlib, it provides a simpler, high-level interface for creating common statistical plots. We will use Seaborn for its ease of use and attractive defaults.

Visualization 1: How many survived? A simple `countplot` is the best

- ✓ way to see the distribution of a categorical variable, like our target `Survived`.

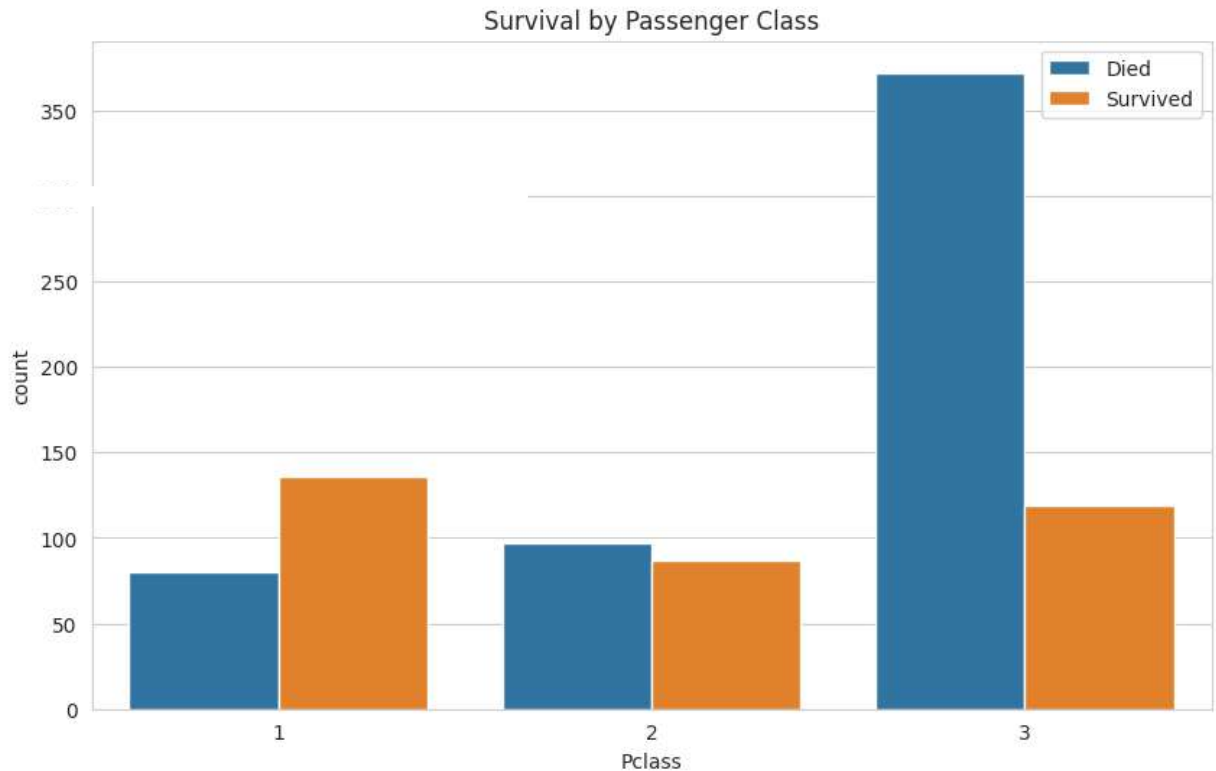
```
sns.set_style('whitegrid') # Sets a nice visual style for our plotsplt.figure(f
```

Visualization 2: Does passenger class matter for survival? Now we want

- ✓ to see if there's a relationship between two variables. We can use the `hue` parameter in `countplot` to split the bars by another category.

```
plt.figure(figsize=(10, 6))
sns.countplot(x='Pclass', hue='Survived', data=df)
plt.title('Survival by Passenger Class')
plt.legend(['Died', 'Survived'])
```

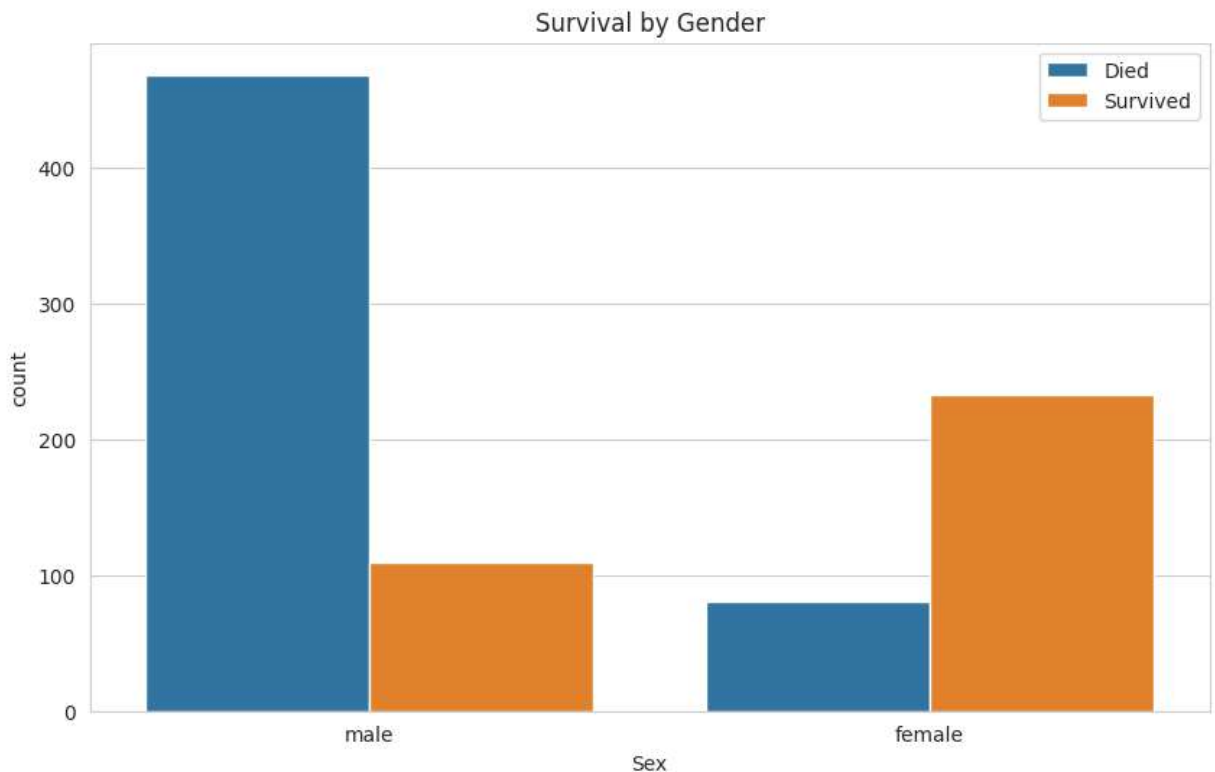
```
plt.show()
print("Insight: This is a very strong pattern. 1st class passengers had a much
```



Insight: This is a very strong pattern. 1st class passengers had a much higher c

✓ Visualization 3: What about gender?The 'women and children first' mantra is famous. Let's see if the data supports it.

```
plt.figure(figsize=(10, 6))
sns.countplot(x='Sex', hue='Survived', data=df)
plt.title('Survival by Gender')
plt.legend(['Died', 'Survived'])
plt.show()
print("Insight: The pattern is undeniable. A much higher proportion of females
```



Insight: The pattern is undeniable. A much higher proportion of females survived

Visualization 4: How does age play a role? For a continuous variable like **Age**, a histogram is a great way to see its distribution.

A FacetGrid allows us to create multiple plots side-by-side to compare distri

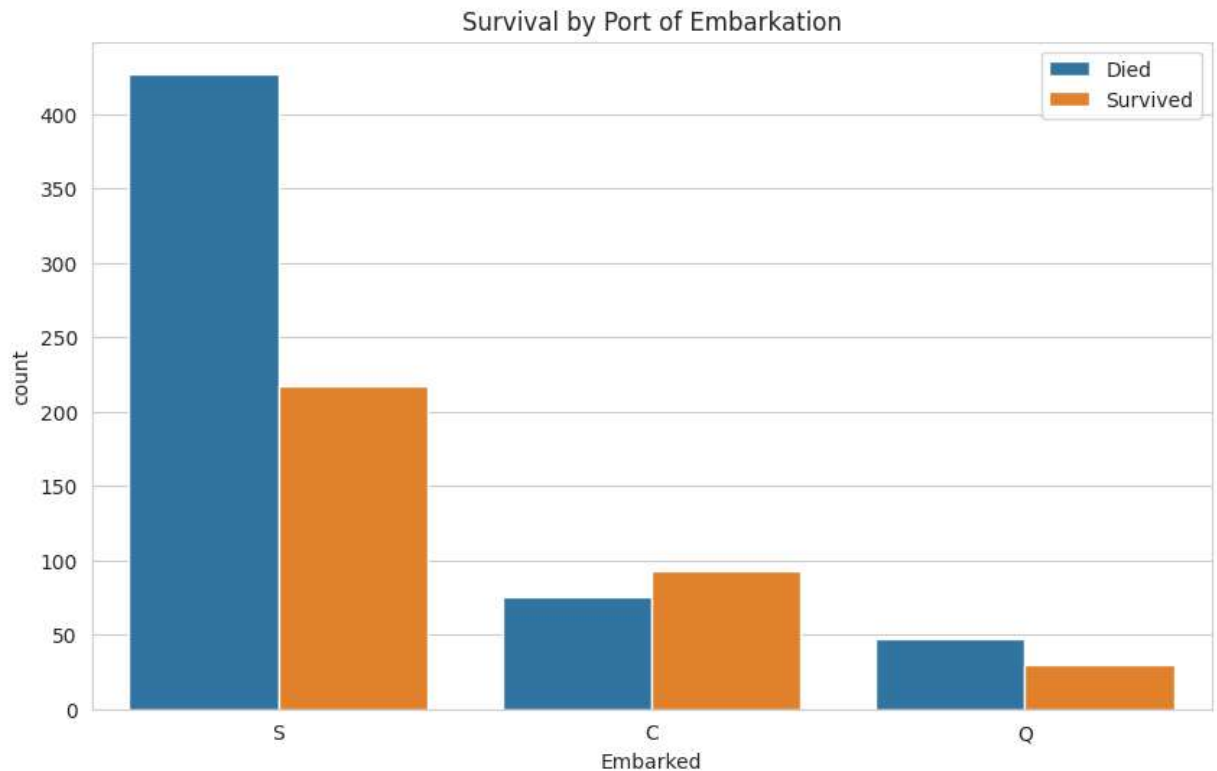
Part 4: Student Experimentation **Instructions:** Create your own visualizations to explore other relationships in the data.

Experiment 1: Port of Embarkation1. The **Embarked** column tells you where a passenger boarded the ship (C = Cherbourg, Q =

Queenstown, S = Southampton).2. Create a **countplot** to see how

survival rates differed by the port of embarkation. Does where they boarded seem to be related to their survival?

```
plt.figure(figsize=(10, 6))
sns.countplot(x='Embarked', hue='Survived', data=df)
plt.title('Survival by Port of Embarkation')
plt.legend(['Died', 'Survived'])
plt.show()
print("Insight: Survival rates appear to vary by embarkation port. Passengers f
```



Insight: Survival rates appear to vary by embarkation port. Passengers from Cher

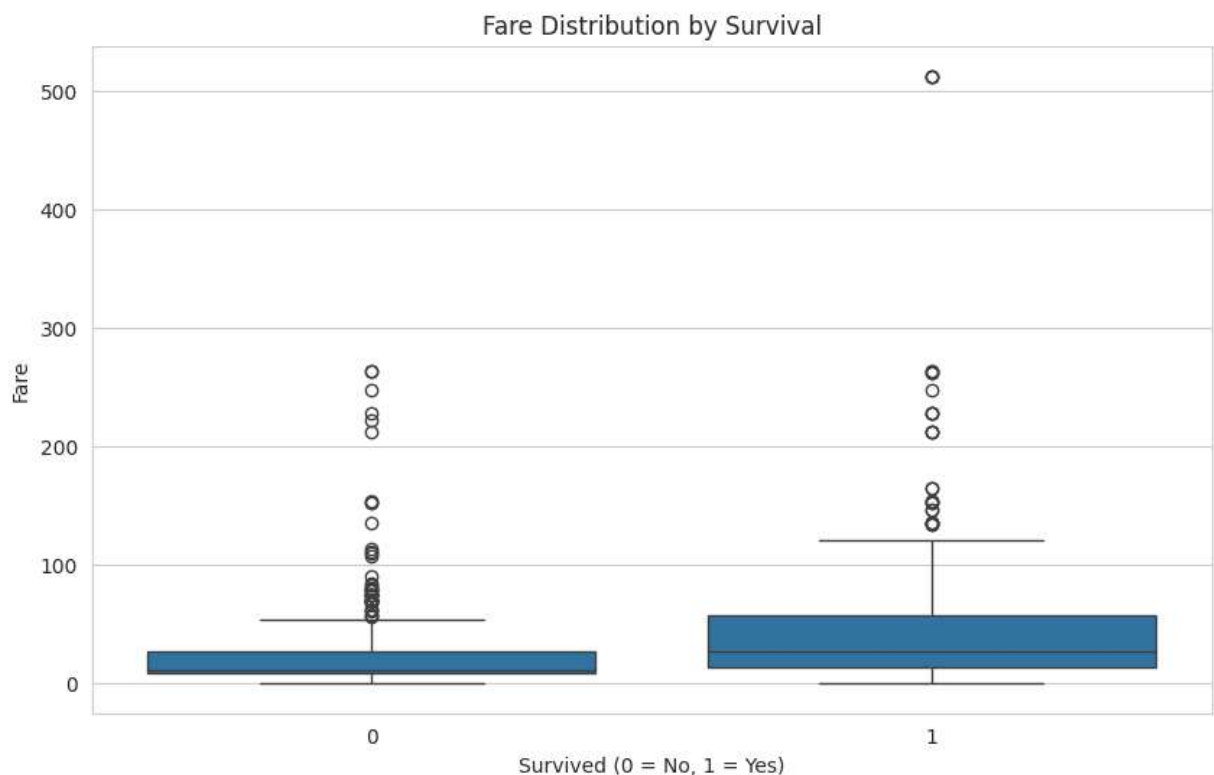
Experiment 2: Fare vs. Survival1. Fare is a continuous numerical

variable.2. A boxplot or violinplot is a great way to see the

✓ distribution of fares for those who survived vs. those who didn't.3.

Create one of these plots to compare the `Fare` distribution by `Survived`. What does it tell you?

```
plt.figure(figsize=(10, 6))
sns.boxplot(x='Survived', y='Fare', data=df)
plt.title('Fare Distribution by Survival')
plt.xlabel('Survived (0 = No, 1 = Yes)')
plt.ylabel('Fare')
plt.show()
print("Insight: Passengers who survived generally paid higher fares, indicating
```



Insight: Passengers who survived generally paid higher fares, indicating a corre



Knowledge CheckInstructions: Answer the following

questions in this markdown cell.1. **What is the primary goal of Exploratory Data Analysis (EDA)?**2. **Based on the plots**

in this lab, what kind of person had the best chance of surviving the Titanic? (Describe them in terms of class, gender, and age).3. Why is it important to visualize data instead of just looking at summary statistics? What can a plot show you that a number like 'mean' or 'count' can't? [ENTER YOUR ANSWERS HERE]

1. **What is the primary goal of Exploratory Data Analysis (EDA)?** The primary goal of EDA is to understand a dataset's main characteristics, identify patterns, detect anomalies, test hypotheses, and extract meaningful insights before formal modeling. It helps in uncovering the 'story' the data tells, identifying errors or missing values, and understanding relationships between variables.
2. **Based on the plots in this lab, what kind of person had the best chance of surviving the Titanic?** Based on the plots, the kind of person who had the best chance of surviving the Titanic was likely a **female, from 1st class, and potentially a young child.**
 - **Class:** 1st class passengers had a significantly higher survival rate compared to 3rd class passengers.
 - **Gender:** A much higher proportion of females survived compared to males.
 - **Age:** There was a noticeable spike in survival for young children, supporting the 'children first' mantra.
3. **Why is it important to visualize data instead of just looking at summary statistics?** What can a plot show you that a number like 'mean' or 'count' can't? Visualizing data is crucial because it allows us to:
 - **Identify patterns and relationships immediately:** Plots make trends and correlations visible that might be hidden in raw numbers or summary statistics. For example, a `countplot` clearly showed the strong relationship between passenger class, gender, and survival.
 - **Detect anomalies and outliers:** Visualizations can easily highlight data points that deviate significantly from the rest of the data, which might be missed by simple means or counts.
 - **Understand distributions:** While a mean gives a central tendency, a histogram or boxplot reveals the shape, spread, and skewness of the data distribution (e.g., the bimodal distribution of age in the survival plots).

- **Communicate insights effectively:** Visuals are much easier for humans to process and understand, making insights more accessible and compelling than tables of numbers.
- **Uncover subtle complexities:** For instance, the FacetGrid showing age distribution by survival status revealed that while adults in the 20-40 range had lower survival, young children had a higher survival rate, a nuance that a single mean age wouldn't convey.